

Foundations of Speech recognition

[Speech recognition]

1.0 Content Block

1970–1990 1971 – DARPA funded a five-year speech recognition research project, Speech Understanding Research, seeking a minimum vocabulary size of 1,000 words. The project considered speech understanding a key to achieving progress in speech recognition, which was later disproved. BBN, IBM, Carnegie Mellon (CMU), and Stanford Research Institute participated. 1972 – The IEEE Acoustics, Speech, and Signal Processing group held a conference in Newton, Massachusetts. 1976 – The first ICASSP was held in Philadelphia, which became a major venue for publishing on speech recognition. During the late 1960s, Leonard Baum developed the mathematics of Markov chains at the Institute for Defense Analysis. A decade later, at CMU, Raj Reddy's students James Baker and Janet M. Baker began using the hidden Markov model (HMM) for speech recognition. James Baker had learned about HMMs while at the Institute for Defense Analysis. HMMs enabled researchers to combine sources of knowledge, such as acoustics, language, and syntax, in a unified probabilistic model. By the mid-1980s, Fred Jelinek's team at IBM created a voice-activated typewriter called Tangora, which could handle a 20,000-word vocabulary. Jelinek's statistical approach placed less emphasis on emulating human brain processes in favor of statistical modelling. (Jelinek's group independently discovered the application of HMMs to speech.) This was controversial among linguists since HMMs are too simplistic to account for many features of human languages. However, the HMM proved to be a highly useful way for modelling speech and replaced dynamic time warping as the dominant speech recognition algorithm in the 1980s.

2.0 Content Block

Speech recognition (automatic speech recognition (ASR), computer speech recognition, or speech-to-text (STT)) is a sub-field of computational linguistics concerned with methods and technologies that translate spoken language into text or other interpretable forms. Speech recognition applications include voice user interfaces, where the user speaks to a device, which "listens" and processes the audio. Common voice applications include interpreting commands for calling, call routing, home automation, and aircraft control. These applications are called direct voice input. Productivity applications include searching audio recordings, creating transcripts, and dictation. Speech recognition can be used to analyse speaker characteristics, such as identifying native language using pronunciation assessment. Voice recognition (speaker identification) refers to identifying the speaker, rather than speech contents. Recognizing the speaker can simplify the task of translating speech in systems trained on a specific person's voice. It can also be used to authenticate the speaker as part of a security process.

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Neural networks became interesting in the late 1980s before beginning to dominate in the 2010s. Neural networks have been used in many aspects of speech recognition, such as phoneme classification, phoneme classification through multi-objective evolutionary algorithms, isolated word recognition, audiovisual speech recognition, audiovisual speaker recognition, and speaker adaptation. Neural networks make fewer explicit assumptions about feature statistical properties than HMMs. When used to estimate the probabilities of a speech segment, neural networks allow natural and efficient discriminative training. However, in spite of their effectiveness in classifying short-time units such as individual phonemes and isolated words, early neural networks were rarely successful for continuous recognition because of their limited ability to model temporal dependencies. One approach was to use neural networks for feature transformation, or dimensionality reduction. However, more recently, LSTM and related recurrent neural networks (RNNs), Time Delay Neural Networks (TDNN's), and transformers demonstrated improved

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where s is the number of substitutions, d is the number of deletions, i is the number of insertions, and n is the number of word references. While computing, the word recognition rate (WRR) is used. The formula is:

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Medical documentation In the health care sector, speech recognition can be implemented in front-end or back-end medical documentation processes. In front-end speech recognition, the provider dictates into a speech-recognition engine, words are displayed as they are recognized, and the speaker is responsible for editing and signing off on the document. In back-end or deferred speech recognition the provider speaks into a digital dictation system, the voice is routed through a speech-recognition machine, and a draft document is routed along with the voice file to an editor, who edits/finalizes the draft and final report. A major issue is that the American Recovery and Reinvestment Act of 2009 (ARRA) provides substantial financial benefits to physicians who utilize an Electronic Health Record (EHR) that complies with "Meaningful Use" standards. These standards require that substantial data be maintained by the EHR. The use of speech recognition is more naturally suited to the generation of narrative text, as part of a radiology/pathology interpretation, progress note or discharge summary; the ergonomic gains of using speech recognition to enter structured discrete data (e.g., numeric values or codes from a list or a controlled vocabulary) are relatively minimal for people who are sighted and who can operate a keyboard and mouse. A more significant issue is that most EHRs have not been expressly tailored to take advantage of voice-recognition capabilities. A large part of a clinician's interaction with EHR involves navigation through the user interface that is heavily dependent on keyboard and mouse; voice-based navigation provides only modest ergonomic benefits. By contrast, many highly customized systems for radiology or pathology dictation implement voice "macros", where the use of certain phrases – e.g., "normal report", will automatically fill in a large number of default values and/or generate boilerplate, which vary with the type of exam – e.g., a chest X-ray vs. a gastrointestinal contrast series for a radiology system.

6.0 Content Block

People with disabilities Speech recognition programs can provide many benefit to those with disabilities. For individuals who are deaf or hard of hearing, speech recognition software can be used to generate captions of conversations. Additionally, individuals who are blind (see blindness and education) or have poor vision can benefit from listening to textual content, as well as garner more functionality from a computer by issuing commands with their voice. The use of voice recognition software, in conjunction with a digital audio recorder and a personal computer running word-processing software, has proven useful for restoring damaged short-term memory capacity in individuals who have suffered a stroke or have undergone a craniotomy. Speech recognition has proven very useful for those who have difficulty using their hands due to causes ranging from mild repetitive stress injuries to disabilities that preclude the use of conventional computer input devices. Individuals with physical disabilities can use voice commands and transcription to navigate electronics hands-free. In fact, people who developed RSI from keyboard use became an early and urgent market for speech recognition. Speech recognition is used in deaf telephony, such as voicemail to text, relay services, and captioned telephone. Individuals with learning disabilities who struggle with thought-to-paper communication may benefit from the software, but the product's fallibility remains a significant consideration for many. In addition, speech to text technology is only an effective aid for those with intellectual disabilities if the proper training and resources are provided (e.g. in the classroom setting). This type of technology can help those with dyslexia, but the potential benefits regarding other disabilities are still in question. Mistakes made by the software hinder its effectiveness, since misheard words take more time to fix.